YANG SUN

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EDUCATION

University of Arizona — **Tucson, USA**
Ph.D. in Astronomy and Astrophysics — *2021 - Present*
Expected Date of Ph.D.: July 2026
Supervisor: Prof. George Rieke
Thesis – *Co-Evolution of SMBH and Galaxies From Cosmic Noon to the Present Day*

Master of Science in Astronomy and Astrophysics — *August 2024*

Beijing Normal University — **Bejing, China**
B.S in Astronomy — *2016 - 2020*

POSITIONS

Graduate Researching Assistant — *2021 - Present*

Graduate Teaching Assistant — *ASTR250 in Fall 2023 and Fall 2025*

RESEARCH INTERESTS

Active Galactic Nuclei; Galaxy Evolution; SF&AGN Feedback

- supermassive black hole and galaxy co-evolution across cosmic time
- AGN-driven multiphase outflow and its impact on host galaxy
- star formation and galaxy assembly history, feedback-induced galaxy quenching

PUBLICATION OVERVIEW

See the full publication list on ADS
ORCID ID: 0000-0001-6561-9443
Total Publications: 44, with citations of 739 (March 2026)
First Author Publications: 7 (6 refereed and 1 non-refereed)

OBSERVING PROPOSALS

ALMA Cycle 12	2025.1.01045.S (7.2h)	PI
ALMA Cycle 12	2025.1.01067.S (45.6h)	co-I
JWST Cycle 4	GO 7335 (14.2h)	co-I

TALKS

* for invited talk

- *MIT Monday Afternoon Talk, Boston, USA — *Nov. 2025*
- *Michigan Extreme Astrophysics Group Talk, Ann Arbor, USA — *Nov. 2025*

- *NOIRLab Flash Talk, Tucson, USA *Sept. 2025*
- CRISOL2025 (flash talk), Toledo, Spain *May 2025*
- JWST/NIRCam Science Team Meeting, Tucson, USA *March 2025*
- Dust and Gas throughout Cosmic Time, Hiroshima, Japan *Dec. 2024*
- JWST/MIRI Meeting, Tucson, USA *Oct. 2024*
- STScI Spring Symposium (flash talk), Baltimore, USA *Apr. 2024*
- Steward Internal Symposium, Tucson, USA *Sept. 2023*
- *Columbia University Seminar, New York, USA *July 2023*
- Galaxy Formation Workshop, Tucson, USA *Apr. 2023*
- IAU Symposium 373, Busan, South Korea *Aug. 2022*

SERVICE AND OUTREACH

- Referee for AAS Journals *2025-present*
- Astronomy Tutoring Offered for Majors and Minors (ATOMM) program in Steward Observatory *2025*
- Teacher for Black Holes Camp in Flandrau Planetarium *2025*
- Co-organizer of Steward Internal Symposium *2025*
- Co-organizer of Prospective Graduate Student Visit in Steward Observatory *2024*
- Member of International Scholars Task Force in Steward Observatory *2021-Present*
- Co-producer of "Numerical Methods" Online Course at Beijing Normal University *2018*
- Volunteer at Beijing Astronomy Planetarium *2017-2019*

HONOR

- University of Arizona Department of Astronomy Graduate Student Award for Outstanding Scholarship *2026*
- "Jingshi" First Prize Scholarship for Excellent Academic Performance *2018-2019*
- Excellent Research Projects for Undergraduates *Sept. 2019*
- "Kuangqiao" Scholarship for Excellent Academic Performance *Nov. 2018*
- Honorable Mention for Mathematical Competition in Modeling *April 2018*
- "Jingshi" Second Prize Scholarship for Excellent Academic Performance *2016-2018*

SKILLS

Programming: Python, FORTRAN, MATLAB
Packages, Software, Tools: LATEX, CASA, TOPCAT, JWST&ALMA Proposal Tools, `BAGPIPES`, `pPXF`, `galfit`, `galight`
Language: English, Chinese

REFERENCES

Prof. George Rieke The University of Arizona ghrieke@gmail.com
Prof. Marcia Rieke The University of Arizona mrieke@gmail.com
Prof. Ann Zabludoff The University of Arizona aiz@arizona.edu

PUBLICATION LIST

Refereed Paper:

As First Author:

1. **Sun, Y.**, Ji, Z., Rieke, G. H., et al., Extreme Neutral Outflow in a non-AGN Quiescent Galaxy at z∼1.3, 2025, arXiv e-prints, arXiv:2504.14682 (ApJ accepted).
2. **Sun, Y.**, Rieke, G. H., Lyu, J., et al., Evolution of the M_*–M_{BH} Relation from z ∼ 6 to the Present Epoch, 2025, ApJ, 983, 165.
3. **Sun, Y.**, Lyu, J., Rieke, G. H., et al., No Evidence for a Significant Evolution of $M_•$–M_* Relation in Massive Galaxies up to z ∼ 4, 2025, ApJ, 978, 98.
4. **Sun, Y.**, Lee, G.-H., Zabludoff, A. I., et al., Evolution of gas flows along the starburst to post-starburst to quiescent galaxy sequence, 2024, MNRAS, 528, 5783-5803.
5. **Sun, Y.**, Yuan, H., Chen, B., Validations and Corrections of the SFD and Planck Reddening Maps Based on LAMOST and Gaia Data, 2022, ApJS, 260, 17.
6. **Sun, Y.**, Deng, D.-S., Yuan, H.-B., Precision of the Chinese Space Station Telescope (CSST) stellar radial velocities, 2021, Research in Astronomy and Astrophysics, 21, 092.

As Significant Contributor:

1. Rieke, G. H., **Sun, Y.**, Lyu, J., et al., Confirming Near- to Mid-infrared Photometrically Identified Obscured AGNs in the JWST Era, 2025, ApJ, 994, 35.
2. Zhu, Y., Bonaventura, N.,**Sun, Y.**, et al., SMILES Data Release II: Probing Galaxy Evolution during Cosmic Noon and Beyond with NIRSpec Medium-Resolution Spectra, 2025, arXiv e-prints, arXiv:2508.12599.
3. Deng, D., **Sun, Y.**, Wang, T., Wang, Y., Jiang, B., Infrared Excess of a Large OB Star Sample, 2022, ApJ, 935, 175.
4. Yuan, H.-B., Deng, D.-S., **Sun, Y.**, A star-based method for precise wavelength calibration of the Chinese Space Station Telescope (CSST) slitless spectroscopic survey, 2021, RAA, 21, 074.
5. Deng, D., **Sun, Y.**, Jian, M., Jiang, B., Yuan, H., Intrinsic Color Indices of Early-type Dwarf Stars, 2020, AJ, 159, 208.

Other Co-author Publications:

1. Shields, T., Rieke, M., Hainline, K., et al., **including Sun, Y.**, JADES: Low Surface Brightness Galaxies at $0.4 < z < 0.8$ in GOODS-S, 2025, arXiv e-prints, arXiv:2511.17738.
2. Scholtz, J., Parlanti, E., Carniani, S., et al., **including Sun, Y.**, Tentative rotation in a galaxy at z ∼ 14 with ALMA, 2025, MNRAS, 544, L113-L120.
3. Stone, M. A., Rieke, G. H., Lyu, J., et al., **including Sun, Y.**, The z = 7.08 Quasar ULAS J1120+0641 May Never Reach a "Normal" Black Hole to Stellar Mass Ratio, 2025, ApJ, 993, 168.
4. D'Eugenio, F., Nelson, E. J., Eisenstein, D. J., et al., **including Sun, Y.**, JADES Dark

Horse: demonstrating high-multiplex observations with JWST/NIRSpec dense-shutter spectroscopy in the JADES Origins Field, 2025, arXiv e-prints, arXiv:2510.11626.

5. Scholtz, J., Carniani, S., Parlanti, E., et al., **including Sun, Y.**, JADES Data Release 4 – Paper II: Data reduction, analysis and emission-line fluxes of the complete spectroscopic sample, 2025, arXiv e-prints, arXiv:2510.01034.

6. Rinaldi, P., Bonaventura, N., Rieke, G. H., et al., **including Sun, Y.**, Not Just a Dot: The Complex UV Morphology and Underlying Properties of Little Red Dots, 2025, ApJ, 992, 71.

7. Woodrum, C., Shivaei, I., Witstok, J., et al., **including Sun, Y.**, JADES: The Star Formation and Dust Attenuation Properties of Galaxies at 3<z<7, 2025, arXiv e-prints, arXiv:2510.00235.

8. D'Eugenio, F., Helton, J. M., Hainline, K., et al., **including Sun, Y.**, JADES and SAPPHIRES: galaxy metamorphosis amidst a huge, luminous emission-line region, 2025, MNRAS, 542, 960-981.

9. Zhu, Y., Egami, E., Fan, X., et al., **including Sun, Y.**, Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$, 2025, arXiv e-prints, arXiv:2509.00153.

10. Rinaldi, P., Rieke, G. H., Wu, Z., et al., **including Sun, Y.**, Beyond the Dot: an LRD-like nucleus at the Heart of an IR-Bright Galaxy and its implications for high-redshift LRDs, 2025, arXiv e-prints, arXiv:2507.17738.

11. Rieke, G. H., Buiten, V. A., Goldberg, C. E., et al., **including Sun, Y.**, Low Accretion Rates in Black Holes in Late-stage Merger Ultraluminous Infrared Galaxies, 2025, ApJ, 988, 17.

12. Zhu, Y., Rieke, M. J., Ji, Z., et al., **including Sun, Y.**, A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images, 2025, ApJ, 986, 162.

13. Zhu, Y., Alberts, S., Lyu, J., et al., **including Sun, Y.**, SMILES: Potentially Higher Ionizing Photon Production Efficiency in Overdense Regions, 2025, ApJ, 986, 18.

14. Lin, X., Fan, X., Sun, F., et al., **including Sun, Y.**, The Large-scale Environments of Low-luminosity AGNs at $3.9 < z < 6$ and Implications for Their Host Dark Matter Halos from a Complete NIRCam Grism Redshift Survey, 2025, arXiv e-prints, arXiv:2505.02896.

15. Zhang, J., Egami, E., Sun, F., et al., **including Sun, Y.**, Abundant Population of Broad H Emitters in the GOODS-N Field Revealed by CONGRESS, FRESCO, and JADES, 2025, arXiv e-prints, arXiv:2505.02895.

16. Lin, X., Egami, E., Sun, F., et al., **including Sun, Y.**, The Luminosity Function and Clustering of H Emitting Galaxies at $z \approx 4 - 6$ from a Complete NIRCam Grism Redshift Survey, 2025, arXiv e-prints, arXiv:2504.08028.

17. Zhu, Y., Rieke, M. J., Ho, L. C., et al., **including Sun, Y.**, Nuclear Winds Drive Cold Gas Outflows on Kiloparsec Scales in Reionization-Era Quasars, 2025, arXiv e-prints, arXiv:2504.02305.

18. D'Eugenio, F., Hainline, K., Arribas, S., et al., **including Sun, Y.**, Forever Blowing Bubbles: What Powers a 24-kpc Ionized Gas Nebula Around a Normal Galaxy at z=6?, 2025, JWST Proposal. Cycle 4, 7335.

19. Hainline, K. N., Maiolino, R., Juodžbalis, I., et al., **including Sun, Y.**, An Investigation into the Selection and Colors of Little Red Dots and Active Galactic Nuclei, 2025,

ApJ, 979, 138.
20. Alberts, S., Lyu, J., Shivaei, I., et al., **including Sun, Y.**, SMILES Initial Data Release: Unveiling the Obscured Universe with MIRI Multiband Imaging, 2024, ApJ, 976, 224.
21. Zhu, Y., Bakx, T. J. L. C., Ikeda, R., et al., **including Sun, Y.**, Discovery of a Unique Close Quasar–DSFG Pair Linked by a [C II] Bridge at z = 5.63, 2024, Research Notes of the American Astronomical Society, 8, 284.
22. Alberts, S., Williams, C. C., Helton, J. M., et al., **including Sun, Y.**, To High Redshift and Low Mass: Exploring the Emergence of Quenched Galaxies and Their Environments at $3 < z < 6$ in the Ultra-deep JADES MIRI F770W Parallel, 2024, ApJ, 975, 85.
23. Shivaei, I., Alberts, S., Florian, M., et al., **including Sun, Y.**, A new census of dust and polycyclic aromatic hydrocarbons at z = 0.7–2 with JWST MIRI, 2024, A&A, 690, A89.
24. Ji, Z., Williams, C. C., Rieke, G. H., et al., **including Sun, Y.**, Extended hot dust emission around the earliest massive quiescent galaxy, 2024, arXiv e-prints, arXiv:2409.17233.
25. Wevers, T., Coughlin, E. R., Pasham, D. R., et al., **including Sun, Y.**, Live to Die Another Day: The Rebrightening of AT 2018fyk as a Repeating Partial Tidal Disruption Event, 2023, ApJL, 942, L33.

Non-Refereed Paper:

1. **Sun, Y.**, Lee, G.-H., Zabludoff, A. I., et al., Evolution of Gas Flows over the Starburst to Post-Starburst to Quiescent Galaxy Sequence, 2023, Resolving the Rise and Fall of Star Formation in Galaxies, 373, 193-198.